

PSMFlows

Successor Measures over a Flow-Indexed Policy Family:
a step-by-step formulation

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Abstract

Zero-shot reinforcement learning methods (FB, PSM) learn a reward-free representation by training successor measures over a family of policies. Both families are defined in *action* space and both contain policies the offline dataset cannot execute: FB’s actor maximises over the whole action set, and PSM’s codebook policies act uniformly at random. On narrow-support data every Bellman backup is then evaluated off the data distribution, and the damage is baked into a representation shared by all downstream tasks.

This report develops the alternative in full. A behaviour-cloned conditional flow $G_\theta(s, \cdot)$ transports a Gaussian prior to the behaviour distribution, and we index policies by the flow’s *noise* variable rather than by actions. The central result (Proposition 3.3) is that this reparameterisation is an exact isometry of behaviour-relative f -divergences: the unknown, state-dependent density ratio π/μ equals the known, state-independent ratio ν/p_0 pointwise. Concentrability with respect to the behaviour policy — the quantity offline RL spends its loss budget controlling — becomes a Gaussian-relative quantity available in closed form.

We then build successor measures over this family in PSM’s affine form, derive the bilinear implementation as a special case of a named assumption, prove the successor feature identity that makes latent Q -iteration fully in-sample, and give a three-rung zero-shot inference ladder with a generalised policy improvement guarantee. Claims we cannot currently prove are stated as conjectures and marked [NOT PROVED].

Contents

1	Setting, notation, and the problem	2
1.1	Setting	2
1.2	The problem, stated as a concentrability problem	3
2	The behaviour flow and its inverse	3
2.1	Conditional flow matching	3
2.2	Regularity, invertibility, and the Jacobian	4
2.3	The flow behaviour model	4
3	The flow-indexed policy family	5
3.1	Three nested classes	5
3.2	The divergence isometry	5
3.3	Transport coherence: when is the indexing family useful?	7

4	Successor measures over the flow family	8
4.1	The two-argument object and its Bellman equation	8
4.2	Affine representation	8
5	The successor-feature identity and latent Q-iteration	9
6	Dataset latents: inverting the flow	10
7	In-sample temporal-difference learning	11
8	Training objective	12
8.1	Derivation	12
8.2	Empirical objective	12
9	Zero-shot inference: a three-rung ladder	12
9.1	Rung 1 — flow-GPI, no test-time training	13
9.2	Rung 2 — in-sample latent Q -iteration	14
9.3	Rung 3 — amortised latent actor	14
10	LatentFlowPSM	14
10.1	Objects	14
10.2	Dataset latents	15
10.3	Task vectors	15
10.4	Losses	15
10.5	Deployment	16
11	Conjectures and open problems	16
	References	17

1 Setting, notation, and the problem

1.1 Setting

We work with a Markov decision process $(\mathcal{S}, \mathcal{A}, P, \gamma)$ with continuous action set $\mathcal{A} \subseteq \mathbb{R}^{d_a}$ and discount $\gamma \in [0, 1)$. *No reward is specified at training time.* We are given a fixed offline dataset

$$\mathcal{D} = \{(s_i, a_i, s'_i)\}_{i=1}^N,$$

generated by an unknown behaviour policy $\mu(a \mid s)$, with state marginal ρ . We write $\Delta(\Omega)$ for the probability measures on Ω and assume throughout that $\mu(\cdot \mid s)$ admits a density with respect to Lebesgue measure on $\mathbb{R}^{d_a}$ for ρ -a.e. s .

Definition 1.1 (Successor measure). For a policy π , an initial state s and a first action a , the discounted successor measure is

$$M^\pi(s, a, X) = \mathbb{E} \left[\sum_{t \geq 1} \gamma^{t-1} \mathbf{1}_{\{s_t \in X\}} \mid s_0 = s, a_0 = a, \pi \right], \quad X \subseteq \mathcal{S} \text{ measurable.}$$

For any reward $r : \mathcal{S} \rightarrow \mathbb{R}$ received on arrival, $Q_r^\pi(s, a) = \int r(x) M^\pi(s, a, dx)$; a single object therefore answers every reward at once, which is the entire premise of zero-shot RL.

Remark 1.2 (Convention). Definition 1.1 sums from $t \geq 1$, so the point mass in the Bellman recursion sits at the *successor* state s' . The alternative convention $\sum_{t \geq 0} \gamma^t \mathbf{1}_{\{s_t \in X\}}$ places it at s and changes Proposition 5.2 to $\psi(s, u, u') = \varphi(s) + \gamma \mathbb{E}_{s'} \psi(s', u', u')$. We use the $t \geq 1$ form because it is what FB uses and what the implemented loss of §8 computes; mixing the two is a common source of off-by-one errors in this literature.

1.2 The problem, stated as a concentrability problem

Every method in this family must, during training, evaluate its measure or critic at successor state–action pairs (s', a') where a' is chosen by the algorithm rather than drawn from the data. The resulting error is governed by a *concentrability coefficient*: writing π_{tr} for the distribution over a' that the training loop actually queries,

$$C(\pi_{\text{tr}}) = \text{ess sup}_{s, a} \frac{\pi_{\text{tr}}(a \mid s)}{\mu(a \mid s)}. \quad (1)$$

Standard offline RL analyses bound the excess value of the learned object by a quantity proportional to $C(\pi_{\text{tr}})$; when C is infinite, no such bound exists, and in practice the model extrapolates confidently into regions it has never seen.

- **FB.** The actor solves $\max_a F(s, a, z)^\top z$ over all of $\mathcal{A}$. The resulting π_{tr} is deterministic and unrelated to μ , so $C = \infty$.
- **PSM.** The basis is trained on hash-codebook policies $\pi_z(s) = \text{UniformSample}(z + \text{hash}(s))$, i.e. $\pi_{\text{tr}} = \text{Unif}(\mathcal{A})$, giving $C = \text{ess sup}_a 1/(|\mathcal{A}| \mu(a \mid s))$. This is finite only when μ has full support and is bounded away from zero — exactly what broad-coverage datasets provide and narrow ones do not.

Both methods therefore work on ExORL-style data and degrade when coverage narrows. The degradation is qualitatively worse than in single-task offline RL, because the corrupted object is a *shared representation*: it fails on all downstream tasks simultaneously.

Two families of fixes exist. Conservatism penalties (CQL, VC-FB/MC-FB) leave the policy family alone and add a loss term that estimates and penalises the divergence in (1). Behavioural priors (FB-CPR) regularise softly toward a dataset. This report takes a third route: *change the policy family so that (1) is computable in closed form and equal to one by construction*.

2 The behaviour flow and its inverse

2.1 Conditional flow matching

Let $p_0 = \mathcal{N}(0, I_{d_a})$ on $\mathbb{R}^{d_a}$ — note $d_u = d_a$ throughout, which is what makes the map below invertible. For $(s, a) \sim \mathcal{D}$, $u_0 \sim p_0$, $u_1 = a$ and $t \sim \text{Unif}[0, 1]$, set $u_t = (1 - t)u_0 + tu_1$, whose velocity along the interpolation path is $\dot{u}_t = u_1 - u_0 = a - u_0$. We fit a conditional vector field $v_\theta : \mathcal{S} \times [0, 1] \times \mathbb{R}^{d_a} \rightarrow \mathbb{R}^{d_a}$ by

$$\mathcal{L}_{\text{flow}}(\theta) = \mathbb{E}_{(s, a) \sim \mathcal{D}, u_0 \sim p_0, t \sim \text{Unif}[0, 1]} \left[\left\| v_\theta(s, t, u_t) - (a - u_0) \right\|_2^2 \right]. \quad (2)$$

Definition 2.1 (Decoder and encoder). $G_\theta(s, u)$ is the value at $t = 1$ of the solution to the initial value problem

$$\frac{du_t}{dt} = v_\theta(s, t, u_t), \quad u_{t=0} = u, \quad (3)$$

and $E_\theta(s, \cdot) = G_\theta(s, \cdot)^{-1}$ is obtained by integrating (3) backward from $t = 1$ to $t = 0$. We write $J(s, u) = \partial G_\theta(s, u) / \partial u \in \mathbb{R}^{d_a \times d_a}$ and $J_t(s, u) = \partial u_t / \partial u$, so $J = J_1$.

2.2 Regularity, invertibility, and the Jacobian

Assumption A1 (Regularity of the field). For ρ -a.e. s : $v_\theta(s, \cdot, \cdot)$ is continuous in t , continuously differentiable in u , and globally Lipschitz in u uniformly in $t \in [0, 1]$, with constant L_s .

Assumption A1 is a statement about the trained network, not about the data; it holds for the standard MLP parameterisations with Lipschitz activations and bounded weights.

Lemma 2.2 (Diffeomorphism and Liouville formula). *Under Assumption A1, for ρ -a.e. s :*

- (i) (3) has a unique solution on $[0, 1]$ for every initial condition, and $G_\theta(s, \cdot)$ is a diffeomorphism of class C^1 from $\mathbb{R}^{d_a}$ onto itself, with inverse $E_\theta(s, \cdot)$ obtained by backward integration;
- (ii) the Jacobian satisfies the variational equation $\dot{J}_t = \partial_u v_\theta(s, t, u_t) J_t$ with $J_0 = I$, and

$$\det J(s, u) = \exp \left(\int_0^1 \text{tr } \partial_u v_\theta(s, t, u_t) dt \right) > 0. \quad (4)$$

Proof. (i) Global Lipschitz continuity in u , uniform in t , gives existence and uniqueness on all of $[0, 1]$ by Picard–Lindelöf, with no finite-time blow-up. The flow map of an ODE is invertible because integrating the same field backward from u_1 recovers u_0 uniquely, again by uniqueness of solutions; surjectivity onto $\mathbb{R}^{d_a}$ follows because backward integration from an arbitrary endpoint is itself well posed and produces a preimage. Continuous differentiability of the flow in its initial condition is the standard smooth-dependence theorem, which applies since $\partial_u v_\theta$ exists and is continuous.

(ii) Differentiating (3) in u and exchanging derivatives (justified by C^1 dependence) gives the variational equation. Liouville’s formula for linear systems states $\frac{d}{dt} \det J_t = \text{tr} (\partial_u v_\theta) \det J_t$; integrating from $\det J_0 = 1$ gives (4), which is a positive exponential and in particular never vanishes. $\square$

Remark 2.3 (Consequence that must not be over-read). Lemma 2.2(i) says the image of $G_\theta(s, \cdot)$ is *all* of $\mathbb{R}^{d_a}$. Consequently $\text{supp} (G_\theta(s, \cdot)_{\#} p_0) = \mathbb{R}^{d_a}$, and the statement “every decoded action lies in $\text{supp } \mu(\cdot \mid s)$ ” is *false* as written. Every guarantee below is therefore a statement about *probability mass* or about *distributions*, never about topological support. See Conjecture 4 for the tension this creates with compactly supported behaviour distributions.

2.3 The flow behaviour model

Definition 2.4. The *flow behaviour model* is $\hat{\mu}_\theta(\cdot \mid s) := G_\theta(s, \cdot)_{\#} p_0$, the pushforward of the prior through the decoder.

Assumption A2 (Exact flow). $\hat{\mu}_\theta(\cdot \mid s) = \mu(\cdot \mid s)$ for ρ -a.e. s .

Assumption A2 is the idealisation under which the clean statements hold. It is the minimiser of (2) in the infinite-capacity, infinite-data limit. It is *not* attainable in the regime this report targets (Conjecture 4), and results are stated with respect to $\hat{\mu}_\theta$ wherever that costs nothing, so that the flow-approximation error is visible rather than hidden.

Lemma 2.5 (Change of variables). *Under Assumption A1, $\hat{\mu}_\theta(\cdot | s)$ has density*

$$\hat{\mu}_\theta(a | s) = \frac{p_0(E_\theta(s, a))}{|\det J(s, E_\theta(s, a))|}. \quad (5)$$

Proof. Immediate from Lemma 2.2 and the change-of-variables formula for the pushforward of a density under a C^1 -diffeomorphism. $\square$

3 The flow-indexed policy family

3.1 Three nested classes

Definition 3.1 (Latent and fixed-index policies). For a state-conditioned latent distribution $\nu(\cdot | s) \in \Delta(\mathbb{R}^{d_a})$, the *latent policy* π_ν acts by drawing $u \sim \nu(\cdot | s)$ and playing $G_\theta(s, u)$, i.e. $\pi_\nu(\cdot | s) = G_\theta(s, \cdot)_\# \nu(\cdot | s)$. Write $\Pi_{\text{lat}} = \{\pi_\nu\}$. Two distinguished subclasses:

- (a) **Fixed-index policies** $\pi_u(s) = G_\theta(s, u)$, one deterministic Markov policy for each $u \in \mathbb{R}^{d_a}$ ($\nu = \delta_u$, the same u at every state). This is the *indexing family* over which the representation is trained.
- (b) **State-wise greedy policies** $\nu(\cdot | s) = \delta_{u^*(s)}$, produced at inference time. These are the policies actually deployed.

Note $\nu = p_0$ recovers $\hat{\mu}_\theta$ itself.

Remark 3.2 (Why Π_{lat} is not the interesting object). Since $G_\theta(s, \cdot)$ is a bijection (Lemma 2.2), every policy on $\mathbb{R}^{d_a}$ equals π_ν for $\nu = (G_\theta(s, \cdot)^{-1})_\# \pi(\cdot | s)$. So Π_{lat} is the set of *all* policies, so any claim that it is exactly the well-covered class is vacuous. The content of the reparameterisation is not *which* policies it can express but *how it measures* them, which is the subject of the next subsection.

3.2 The divergence isometry

Proposition 3.3 (Behaviour-relative divergences are Gaussian-relative divergences). *Let Assumption A1 hold and let $\nu(\cdot | s)$ admit a density. Then for ρ -a.e. s :*

- (i) (pointwise ratio identity) *for every $a \in \mathbb{R}^{d_a}$, writing $u = E_\theta(s, a)$,*

$$\frac{\pi_\nu(a | s)}{\hat{\mu}_\theta(a | s)} = \frac{\nu(u | s)}{p_0(u)}; \quad (6)$$

- (ii) (isometry) *for every convex f with $f(1) = 0$,*

$$D_f(\pi_\nu(\cdot | s) \| \hat{\mu}_\theta(\cdot | s)) = D_f(\nu(\cdot | s) \| p_0). \quad (7)$$

Under Assumption A2 one may replace $\hat{\mu}_\theta$ by μ throughout.

Proof. (i) By Lemma 2.5 applied to ν and to p_0 under the same map $G_\theta(s, \cdot)$,

$$\pi_\nu(a \mid s) = \frac{\nu(u \mid s)}{|\det J(s, u)|}, \quad \hat{\mu}_\theta(a \mid s) = \frac{p_0(u)}{|\det J(s, u)|}, \quad u = E_\theta(s, a).$$

Both densities are evaluated at the same point a , hence at the same preimage u , hence carry the *same* Jacobian factor. Taking the ratio cancels it, and the factor is nonzero by (4), so the division is legitimate.

(ii) Substituting $a = G_\theta(s, u)$ and using $\hat{\mu}_\theta(da \mid s) = p_0(du)$ (definition of pushforward),

$$D_f(\pi_\nu \parallel \hat{\mu}_\theta) = \int f\left(\frac{\pi_\nu(a \mid s)}{\hat{\mu}_\theta(a \mid s)}\right) \hat{\mu}_\theta(da \mid s) \stackrel{(i)}{=} \int f\left(\frac{\nu(u \mid s)}{p_0(u)}\right) p_0(du) = D_f(\nu \parallel p_0). \quad \square$$

Equation (7) is the load-bearing statement of this report, so it is worth saying plainly what it buys. The left-hand side involves μ , which is unknown, is a different distribution at every state, and has geometry nobody can characterise. The right-hand side involves only $p_0 = \mathcal{N}(0, I)$ and the latent distribution we choose. Every behaviour-regularised objective of the form $\max_\pi J(\pi)$ s.t. $D_f(\pi \parallel \mu) \leq \epsilon$ — the shape that CQL, BRAC, VC-FB and most of offline RL instantiate — becomes $\max_\nu J(\pi_\nu)$ s.t. $D_f(\nu \parallel p_0) \leq \epsilon$, whose constraint is available in closed form for Gaussian ν and requires *no density model of the behaviour policy at all*. In particular, for $f(x) = x \log x$, $\text{KL}(\pi_\nu(\cdot \mid s) \parallel \mu(\cdot \mid s)) = \text{KL}(\nu(\cdot \mid s) \parallel \mathcal{N}(0, I))$, exactly, at every state.

Corollary 3.4 (Mass conversion). *For every measurable $U \subseteq \mathbb{R}^{d_a}$ and ρ -a.e. s ,*

$$\hat{\mu}_\theta(G_\theta(s, U) \mid s) = p_0(U). \quad (8)$$

In particular, for $U_\delta := \{u : \|u\|^2 \leq \chi_{d_a}^2(1 - \delta)\}$ we have $\hat{\mu}_\theta(G_\theta(s, U_\delta) \mid s) = 1 - \delta$ simultaneously at every state, where $\chi_{d_a}^2(1 - \delta)$ is the $(1 - \delta)$ -quantile of the chi-squared distribution with d_a degrees of freedom.

Proof. $\hat{\mu}_\theta(G_\theta(s, U) \mid s) = p_0(G_\theta(s, \cdot)^{-1}(G_\theta(s, U))) = p_0(U)$ by the definition of pushforward and bijectivity (Lemma 2.2). The second claim is $\|u\|^2 \sim \chi_{d_a}^2$ under p_0 . $\square$

This is the precise version of “the flow converts a hard constraint into an easy one”. The constraint “ a carries at least $1 - \delta$ of the behaviour mass at s ” is state-dependent with unknown geometry; its latent image is a *single Euclidean ball, the same at every state, known before training begins*.

Corollary 3.5 (Bounded concentrability for stochastic latent policies). *If $\|\text{d}\nu(\cdot \mid s)/\text{d}p_0\|_\infty \leq \kappa$ for ρ -a.e. s , then $C(\pi_\nu) \leq \kappa$ in the sense of (1) with μ replaced by $\hat{\mu}_\theta$. Taking $\nu = p_0$ gives $C = 1$.*

Proof. Immediate from (6) and the definition (1). $\square$

Remark 3.6 (The two regimes must not be conflated). Corollaries 3.4 and 3.5 govern different objects and the distinction is load-bearing.

- A *deterministic* policy — both the fixed-index π_u of Definition 3.1(a) and the deployed greedy policy of Definition 3.1(b) — has $\nu = \delta_u \not\ll p_0$ and therefore *infinite* f -divergence from μ . Corollary 3.5 says nothing about it. Its governing guarantee is the mass statement, Corollary 3.4: confine u to U_δ and the action carries $1 - \delta$ of the behaviour mass.

- *At training time* the fixed-index family is never used in isolation: the loss of §8 averages over $u' \sim p_0$, so the marginal law of the bootstrap actions is exactly $\hat{\mu}_\theta(\cdot \mid s')$ and the operator’s concentrability is 1. That is the content of Proposition 7.2, and it is a statement about the *aggregate* training distribution, not about any individual $\pi_{u'}$.

Corollary 3.7 (Density-adapted resolution). *For ρ -a.e. s and every u ,*

$$|\det J(s, u)| = \frac{p_0(u)}{\hat{\mu}_\theta(G_\theta(s, u) \mid s)}. \quad (9)$$

Proof. Rearrange (5) at $a = G_\theta(s, u)$. □

Equation (9) makes the “implicit trust region” claim quantitative: the local volume expansion of the decoder is exactly the ratio of prior density to behaviour density. Where the data is dense the map contracts, so a fixed step in u moves the action little; across low-density gaps it expands, so the same step jumps over the gap. Any optimiser working at fixed resolution in u therefore searches finely inside data modes and coarsely between them, without being told where the modes are.

3.3 Transport coherence: when is the indexing family useful?

Nothing so far guarantees that a *fixed* u selects behaviourally consistent actions across states. If it does not, the family $\{\pi_u\}$ is a collection of incoherent policies and, while every result above still holds, the representation has little structure to exploit. We make this measurable.

Definition 3.8 (Transport coherence). Let $\{C_k(s)\}_{k=1}^m$ be a measurable partition of $\mathbb{R}^{d_a}$ for each s (in practice: the cells of a mode decomposition of $\hat{\mu}_\theta(\cdot \mid s)$, e.g. mixture components or k -means cells of the dataset actions). The decoder G_θ is ε -coherent with respect to $\{C_k\}$ if there exists a labelling $\kappa : \mathbb{R}^{d_a} \rightarrow \{1, \dots, m\}$ with

$$(\rho \otimes p_0) \left\{ (s, u) : G_\theta(s, u) \notin C_{\kappa(u)}(s) \right\} \leq \varepsilon. \quad (10)$$

That is, a fixed latent lands in the same-labelled mode at all but an ε -fraction of state–latent pairs.

Assumption A3 (Coherence). G_θ is ε -coherent with $\varepsilon \ll (m - 1)/m$.

Remark 3.9 (What Assumption A3 is and is not needed for). Assumption A3 is used by *none* of Propositions 3.3, 5.2, 7.2 or 9.1: soundness of the representation and of the backups does not depend on it. It is a hypothesis about *statistical efficiency*. If ε approaches the chance level $(m - 1)/m$, then a fixed u selects modes essentially at random across states, the successor measures $\{M^{\pi_u}\}$ all concentrate near $M^{\hat{\mu}_\theta}$, the family spans a low-dimensional and uninformative set, and the basis dimension d_z needed to distinguish useful policies grows. The quantity ε in (10) is directly estimable from rollouts, so Assumption A3 is falsifiable before any representation is trained. If it fails, the mitigation is to re-index the family — learn $\nu_\eta(u \mid s, z)$ for a low-dimensional z with a mutual-information term — which preserves every result in §3 because those results hold for arbitrary ν .

4 Successor measures over the flow family

4.1 The two-argument object and its Bellman equation

Definition 4.1. For $u, u' \in \mathbb{R}^{d_a}$, let $M^{u \rightarrow u'}(s, \cdot)$ denote the successor measure from s when the first action is $G_\theta(s, u)$ and the continuation policy is $\pi_{u'}$:

$$M^{u \rightarrow u'}(s, \cdot) = M^{\pi_{u'}}(s, G_\theta(s, u), \cdot)$$

in the notation of Definition 1.1. We write $m(s, u, u', x)$ for its density with respect to ρ .

Lemma 4.2 (Bellman equation; diagonal closure). *For all s, u, u' and measurable X ,*

$$M^{u \rightarrow u'}(s, X) = \mathbb{E}_{s' \sim P(\cdot | s, G_\theta(s, u))} [\mathbf{1}_{\{s' \in X\}} + \gamma M^{u' \rightarrow u'}(s', X)]. \quad (11)$$

Proof. Split the sum of Definition 1.1 at $t = 1$ and apply the Markov property. After the first transition the agent follows $\pi_{u'}$, so its first action at s' is $G_\theta(s', u')$ and its continuation is again $\pi_{u'}$; both arguments of the successor object are therefore u' . $\square$

The recursion *closes on the diagonal*: only $M^{u' \rightarrow u'}$ appears on the right. The two-argument object is exactly the Q -like one-step extension of the V -like diagonal object, in the same relation as (Q^π, V^π) , and this is what makes a single learned network sufficient.

4.2 Affine representation

Assumption A4 (Affine successor-measure form). There exist a bias b , a basis Φ taking values in $\mathbb{R}^{d_z}$, and coordinates $w^{u'} \in \mathbb{R}^{d_z}$ such that

$$m(s, u, u', x) = b(s, u, x) + \Phi(s, u, x)^\top w^{u'} \quad (12)$$

for all s, u, u', x , where b and Φ *do not depend on the policy index u'* .

Assumption A4 is PSM's affine decomposition instantiated on the flow family: the policy enters the successor measure only through a finite-dimensional coordinate. PSM proves the corresponding statement for finite MDPs over all policies; the version we need — restricted to $\{\pi_{u'}\}$ — is Conjecture 2.

Substituting (12) into (11) gives the constraint the learned objects must satisfy:

$$b(s, u, x) + \Phi(s, u, x)^\top w^{u'} = \mathbb{E}_{s'} \left[\frac{\delta_{s'}(dx)}{\rho(dx)} + \gamma (b(s', u', x) + \Phi(s', u', x)^\top w^{u'}) \right], \quad (13)$$

with $s' \sim P(\cdot | s, G_\theta(s, u))$. Note the asymmetry: the left side is evaluated at index u , the right at u' , which is precisely the diagonal closure of Lemma 4.2.

Assumption A5 (Factorised basis). There exist $\varphi : \mathcal{S} \rightarrow \mathbb{R}^d$, $A : \mathcal{S} \times \mathbb{R}^{d_a} \rightarrow \mathbb{R}^{d_z \times d}$ and $\beta : \mathcal{S} \times \mathbb{R}^{d_a} \rightarrow \mathbb{R}^d$ with

$$\Phi(s, u, x) = A(s, u) \varphi(x), \quad b(s, u, x) = \beta(s, u)^\top \varphi(x).$$

Proposition 4.3 (The bilinear form is a special case). *Under Assumptions A4 and A5,*

$$m(s, u, u', x) = \psi(s, u, u')^\top \varphi(x), \quad \psi(s, u, u') = A(s, u)^\top w^{u'} + \beta(s, u), \quad (14)$$

so m is bilinear in (ψ, φ) and ψ is affine in the policy coordinate $w^{u'}$.

Proof. $b + \Phi^\top w^{u'} = \beta(s, u)^\top \varphi(x) + (A(s, u)\varphi(x))^\top w^{u'} = (\beta(s, u) + A(s, u)^\top w^{u'})^\top \varphi(x)$. $\square$

Remark 4.4 (What the implementation keeps and what it gives up). The implemented agent parameterises $\psi(s, u, u')$ as a free network, i.e. it adopts (14) but *not* the affineness of ψ in $w^{u'}$. The consequences are worth stating explicitly.

	affine (12)	factorised (14)	free ψ (implemented)
policy-independent basis	Φ, b	φ	φ
policy index	$w^{u'}$	$w^{u'}$	absorbed into ψ
closed-form reward inference	yes	yes	yes
constrained (LP) inference over w	yes	yes	no
GPI guarantee (Prop. 9.1)	yes	yes	yes

The constrained-optimisation inference of full PSM works *only because* m is linear in the task coordinate; a free ψ forfeits it and retains closed-form w only. This is a deliberate trade, not an oversight.

5 The successor-feature identity and latent Q -iteration

Definition 5.1. The successor features of the flow family are $\psi(s, u, u') = \int \varphi(x) M^{u \rightarrow u'}(s, dx)$.

Proposition 5.2 (Successor-feature Bellman identity). *For all s, u, u' ,*

$$\psi(s, u, u') = \mathbb{E}_{s' \sim P(\cdot | s, G_\theta(s, u))} [\varphi(s') + \gamma \psi(s', u', u')]. \quad (15)$$

Proof. Integrate (11) against φ and exchange integrals (Fubini; φ bounded, M finite since $\gamma < 1$). $\square$

Corollary 5.3 (Reward identity: the draft conjecture, settled). *For any $w \in \mathbb{R}^d$, with $\hat{r}_w(x) := \varphi(x)^\top w$,*

$$\mathbb{E}_{s'} [(\psi(s, u, u') - \gamma \psi(s', u', u'))^\top w] = \mathbb{E}_{s'} [\hat{r}_w(s')], \quad (16)$$

which is independent of the continuation index u' — and indeed of u except through the transition kernel.

Proof. Take inner products of (15) with w and rearrange. $\square$

An earlier draft of this work conjectured (16) with the remark “if I’m not completely mistaken”. The conjecture is correct, it is nothing more than the successor-feature Bellman identity, and it holds for *every* u' rather than approximately. Two consequences follow.

Corollary 5.4 (Closed-form reward inference). *If $\mathbb{E}_{x \sim \rho} [\varphi(x)\varphi(x)^\top] = I_d$, then the least-squares projection of a reward r onto the basis is $w = \mathbb{E}_{x \sim \rho} [r(x)\varphi(x)]$.*

Proof. Minimising $\mathbb{E}_\rho [(r - \varphi^\top w)^2]$ gives the normal equations $\mathbb{E}_\rho [\varphi\varphi^\top]w = \mathbb{E}_\rho [r\varphi]$; the assumed orthonormality reduces the left side to w . $\square$

Corollary 5.4 explains the role of the orthonormality regulariser in §8: it is not a stabilisation heuristic but the condition under which the closed-form w is the correct least-squares answer.

Corollary 5.5 (Latent Q -iteration). *Fix w and a latent action set $\mathcal{U} \subseteq \mathbb{R}^{d_a}$. The optimal value over state-wise latent selection satisfies*

$$Q_w^*(s, u) = \mathbb{E}_{s' \sim P(\cdot | s, G_\theta(s, u))} \left[\hat{r}_w(s') + \gamma \max_{u' \in \mathcal{U}} Q_w^*(s', u') \right]. \quad (17)$$

Proof. (17) is the Bellman optimality equation of the MDP whose action set is $\mathcal{U}$, whose transition kernel is $P(\cdot | s, G_\theta(s, u))$, and whose reward is $\hat{r}_w$ received on arrival. The right-hand side defines a γ -contraction in $\|\cdot\|_\infty$ by the standard argument, so the fixed point exists and is unique. $\square$

This is an ordinary Q -learning problem posed in the latent variable. The maximisation is over u' , and every u' decodes to an action drawn from the flow behaviour model, so no out-of-distribution action query appears anywhere in the loop. Taking $\mathcal{U} = U_\delta$ gives, by Corollary 3.4, an explicit $1 - \delta$ behaviour-mass guarantee on the resulting greedy policy at every state.

6 Dataset latents: inverting the flow

The updates of §8 require triples (s, u, s') with $s' \sim P(\cdot | s, G_\theta(s, u))$, but the dataset supplies (s, a, s') . We need a latent u with $G_\theta(s, u) = a$. By Lemma 2.2 there is exactly one, namely $u = E_\theta(s, a)$, computable by backward integration.

Lemma 6.1 (Inverted latents are prior-distributed). *Under Assumptions A1 and A2, if $(s, a) \sim \mathcal{D}$ then $E_\theta(s, a) \sim p_0$.*

Proof. $a \sim \mu(\cdot | s) = \hat{\mu}_\theta(\cdot | s) = G_\theta(s, \cdot)_{\#} p_0$; pushing forward through $E_\theta(s, \cdot) = G_\theta(s, \cdot)^{-1}$ returns p_0 . $\square$

Lemma 6.1 gives a direct test of Assumption A2: comparing the empirical distribution of $\|E_\theta(s, a)\|^2$ against $\chi_{d_a}^2$ tests exactness of the flow, not merely numerical health of the integrator. If inverted latents are atypical under p_0 , then the latents seen at training time and the samples $u' \sim p_0$ used at evaluation time come from different distributions, and the representation is being trained and queried on different families.

Because the trained flow is imperfect, we replace the point preimage by the ε -relaxed posterior

$$q_\alpha(u | s, a) \propto p_0(u) \exp(-\alpha \|G_\theta(s, u) - a\|^2), \quad \alpha > 0. \quad (18)$$

Proposition 6.2 (Laplace approximation of the preimage posterior). *Let $u^* = E_\theta(s, a)$ and $J = J(s, u^*)$. The second-order expansion of (18) about u^* is $\mathcal{N}(m_\alpha, \Sigma_\alpha)$ with*

$$\Sigma_\alpha = (I + 2\alpha J^\top J)^{-1}, \quad m_\alpha = 2\alpha \Sigma_\alpha J^\top J u^*. \quad (19)$$

As $\alpha \rightarrow \infty$, $m_\alpha \rightarrow u^*$ and $\Sigma_\alpha \rightarrow (2\alpha)^{-1}(J^\top J)^{-1}$.

Proof. Write $\delta = u - u^*$. A first-order expansion of the decoder gives $G_\theta(s, u) - a = J\delta + O(\|\delta\|^2)$, so the likelihood term is $\exp(-\alpha \delta^\top J^\top J \delta) = \exp(-\frac{1}{2} \delta^\top (2\alpha J^\top J) \delta)$, a Gaussian factor with precision $2\alpha J^\top J$. The prior $p_0 = \mathcal{N}(0, I)$ has precision I . Multiplying Gaussians adds precisions, giving $\Sigma_\alpha^{-1} = I + 2\alpha J^\top J$, and the mean is Σ_α times the sum of precision-weighted means, $2\alpha J^\top J u^* + I \cdot 0$. For the limit, $(I + 2\alpha J^\top J)^{-1} 2\alpha J^\top J \rightarrow I$ as $\alpha \rightarrow \infty$ whenever J is nonsingular, which holds by (4). $\square$

Remark 6.3 (The posterior spread is the local geometry). Pushing $\Sigma_\alpha \rightarrow (2\alpha)^{-1}(J^\top J)^{-1}$ forward through the linearised decoder gives covariance $J\Sigma_\alpha J^\top \rightarrow (2\alpha)^{-1}I$ in *action* space. So (18) is an isotropic ball of radius $\sim (2\alpha)^{-1/2}$ around a , pulled back through the flow. Its anisotropy in latent space is not a modelling choice but the inverse Jacobian geometry: the posterior is wide in u exactly where the decoder contracts, i.e. where the data is dense (Corollary 3.7). The Gaussian mixture fitted by EM in the implementation is a multi-modal refinement of this Laplace object, not an unrelated heuristic.

7 In-sample temporal-difference learning

Definition 7.1 (The PSMFlows TD operator). For a candidate m , define

$$(\mathcal{T}m)(s, u, u', x) = \mathbb{E}_{s' \sim P(\cdot | s, G_\theta(s, u))} \left[\frac{\delta_{s'}(\mathrm{d}x)}{\rho(\mathrm{d}x)} + \gamma m(s', u', u', x) \right], \quad (20)$$

where at s' the bootstrap action is $G_\theta(s', u')$ with $u' \sim p_0$.

Proposition 7.2 (In-sample property). *Under Assumption A2, if $s' \sim \rho$ and $u' \sim p_0$ independently, then the bootstrap action satisfies $G_\theta(s', u') \sim \mu(\cdot | s')$. Consequently the joint law of the pairs (s', a') at which $\mathcal{T}$ evaluates m is exactly the data distribution $\rho \otimes \mu$, and the concentrability coefficient (1) of the operator equals 1.*

Proof. $G_\theta(s', \cdot)_{\#} p_0 = \hat{\mu}_\theta(\cdot | s') = \mu(\cdot | s')$ by Definition 2.4 and Assumption A2. Independence of u' and s' gives the joint law, and (1) is then a ratio of identical densities. $\square$

Proposition 7.2 is the formal separation from FB and PSM. Placing it beside §1.2:

Method	bootstrap action distribution	$C(\pi_{\text{tr}})$
FB	$\delta_{\arg \max_a F(s', a, z)^\top z}$	∞
PSM	$\text{Unif}(\mathcal{A})$	$\text{ess sup}_a 1/(\mathcal{A} \mu(a s'))$
PSMFlows	$G_\theta(s', \cdot)_{\#} p_0 = \hat{\mu}_\theta(\cdot s')$	1

Proposition 7.3 (Contraction and fixed-point error). *$\mathcal{T}$ is a γ -contraction in $\|\cdot\|_\infty$ on bounded measurable functions, with unique fixed point $m(s, u, u', \cdot) = M^{u \rightarrow u'}(s, \cdot)/\rho$. If a learned $\hat{m}$ satisfies $\|\hat{m} - \mathcal{T}\hat{m}\|_\infty \leq \epsilon_{\text{TD}}$ then*

$$\|\hat{m} - m\|_\infty \leq \frac{\epsilon_{\text{TD}}}{1 - \gamma}. \quad (21)$$

Proof. For bounded m_1, m_2 , $\|\mathcal{T}m_1 - \mathcal{T}m_2\|_\infty \leq \gamma\|m_1 - m_2\|_\infty$ since the point-mass term cancels and the expectation is an average. Lemma 4.2 identifies the fixed point. For (21), $\|\hat{m} - m\| \leq \|\hat{m} - \mathcal{T}\hat{m}\| + \|\mathcal{T}\hat{m} - \mathcal{T}m\| \leq \epsilon_{\text{TD}} + \gamma\|\hat{m} - m\|$. $\square$

Remark 7.4 (What is and is not new here). Proposition 7.3 is entirely standard; contraction is never the difficult part. What Proposition 7.2 adds is that ϵ_{TD} is estimable *from the dataset alone*: the Bellman residual is evaluated at pairs drawn from $\rho \otimes \mu$, which is what the data provides. For FB and PSM the corresponding residual is evaluated at pairs the data does not contain, so the empirical residual is not an estimate of ϵ_{TD} at all, and the bound requires a concentrability factor over $\mathcal{A}$ that is unbounded on narrow data. A sample-complexity statement making this quantitative in the continuous case is Conjecture 3; in the finite-state, finite-latent-codebook case it follows from the standard in-sample offline RL analyses with state-coverage C_ρ in place of state-action coverage.

8 Training objective

8.1 Derivation

We fit m by minimising the $L^2(\rho)$ distance to its Bellman target, with a stop-gradient target $\bar{m}$:

$$\mathcal{L}(m) = \mathbb{E}_{x \sim \rho} \left[\left(m(s, u, u', x) - (\mathcal{T}\bar{m})(s, u, u', x) \right)^2 \right].$$

Expanding the square and using $\mathbb{E}_{x \sim \rho} \left[m(\cdot, x) \frac{\delta_{s'}(\mathrm{d}x)}{\rho(\mathrm{d}x)} \right] = m(\cdot, s')$, all terms not depending on m are constants, leaving (writing $\stackrel{c}{=}$ for equality up to an additive constant independent of m)

$$\mathcal{L}(m) \stackrel{c}{=} \mathbb{E}_{x \sim \rho} \left[\left(m(s, u, u', x) - \gamma \bar{m}(s', u', u', x) \right)^2 \right] - 2 \mathbb{E}_{s'} [m(s, u, u', s')]. \quad (22)$$

The first term is an ordinary squared Bellman residual over a random “negative” state $x \sim \rho$; the second is a “positive” term rewarding mass at the observed successor. This is the norm-squared measure-matching loss of FB and PSM, here instantiated on the flow family.

8.2 Empirical objective

Let $\{(s_i, a_i, s'_i)\}_{i=1}^B$ be a batch, $u_i \sim q_\alpha(\cdot \mid s_i, a_i)$ the inverted latents of §6, $u' \sim p_0$ a fresh index per batch element, and $\bar{\psi}, \bar{\varphi}$ target networks. Using the bilinear form (14) and drawing negatives $x = s'_j$ from other batch elements:

$$\boxed{\begin{aligned} \mathcal{L}_{\text{SM}} = & \frac{1}{B^2} \sum_{i,j} \left(\psi(s_i, u_i, u')^\top \varphi(s'_j) - \gamma \bar{\psi}(s'_i, u', u')^\top \bar{\varphi}(s'_j) \right)^2 \\ & - \frac{2}{B} \sum_i \psi(s_i, u_i, u')^\top \varphi(s'_i) + \lambda_\perp \mathcal{L}_{\text{ortho}}(\varphi), \end{aligned}} \quad (23)$$

with $\mathcal{L}_{\text{ortho}}(\varphi) = \left\| \frac{1}{B} \sum_i \varphi(s_i) \varphi(s_i)^\top - I_d \right\|_F^2$.

Remark 8.1 (The load-bearing line). In (23) the bootstrap term evaluates $\bar{\psi}$ at the successor state s'_i with continuation index u' , whose decoded action is $G_\theta(s'_i, u')$ — a sample from $\hat{\mu}_\theta(\cdot \mid s'_i)$ by Proposition 7.2. *Every action the backup ever evaluates is a flow decode.* The corresponding line in PSM evaluates a uniform random action; in FB, an unconstrained actor maximiser.

Remark 8.2 (Role of $\mathcal{L}_{\text{ortho}}$). By Corollary 5.4, $\mathcal{L}_{\text{ortho}}$ enforces exactly the condition $\mathbb{E}_\rho[\varphi\varphi^\top] = I$ under which the closed-form reward inference $w = \mathbb{E}_\mathcal{D}[r\varphi]$ is the least-squares projection. It is part of the specification, not a regulariser tuned for stability.

Freezing θ after Stage A means the representation is trained against a *stationary* policy family. FB trains against a moving actor, so its measure targets chase a family that changes underneath them.

9 Zero-shot inference: a three-rung ladder

Given a reward r , set $w = \mathbb{E}_\mathcal{D}[r(x)\varphi(x)]$ (Corollary 5.4). The three rungs trade computation at task time against optimality.

Algorithm 1 PSMFlows pretraining (reward-free)

Require: dataset $\mathcal{D}$, prior p_0 , discount γ , sharpness α

- 1: **Stage A.** Fit v_θ by (2); **freeze** θ .
- 2: **Stage B.** For each $(s_i, a_i) \in \mathcal{D}$ precompute $u_i^* = E_\theta(s_i, a_i)$ and $J(s_i, u_i^*)$ by backward ODE; fit $q_\alpha(\cdot \mid s_i, a_i)$ via Proposition 6.2 (EM refinement optional). Cache.
- 3: **Stage C. repeat**
- 4: sample batch $\{(s_i, a_i, s'_i)\}$, draw $u_i \sim q_\alpha(\cdot \mid s_i, a_i)$, draw $u' \sim p_0$
- 5: update (ψ, φ) by gradient descent on (23); update targets $\bar{\psi}, \bar{\varphi}$
- 6: **until** converged

Ensure: basis φ , successor features ψ , frozen decoder G_θ

Algorithm 2 Rung 1: flow-GPI action selection

Require: state s , task code w , budget K , level δ

- 1: draw $u'_1, \dots, u'_K \sim p_0$; draw $u_1, \dots, u_K \sim p_0$ $\triangleright$ each lies in U_δ w.p. $1 - \delta$ (Cor. 3.4)
 - 2: $(\hat{i}, \hat{j}) \leftarrow \arg \max_{i,j} \psi(s, u_i, u'_j)^\top w$
 - 3: optionally refine $u_{\hat{i}}$ by a few gradient steps on $u \mapsto \psi(s, u, u'_{\hat{j}})^\top w$, **projected onto** U_δ
 - 4: **return** $G_\theta(s, u_{\hat{i}})$
-

9.1 Rung 1 — flow-GPI, no test-time training

Proposition 9.1 (GPI over the flow continuum). *Let $\Lambda_K = \{u'_1, \dots, u'_K\}$ and let π^{GPI} select at each s the action $G_\theta(s, \hat{u}(s))$ with $\hat{u}(s) \in \arg \max_{u \in U_\delta} \max_{u' \in \Lambda_K} \psi(s, u, u')^\top w$. Suppose*

$$|\psi(s, u, u')^\top w - Q_r^{\pi_{u'}}(s, G_\theta(s, u))| \leq \epsilon \quad \text{for all } s, u, u' \in \Lambda_K.$$

Then for all s, a ,

$$Q_r^{\pi^{\text{GPI}}}(s, a) \geq \max_{u' \in \Lambda_K} Q_r^{\pi_{u'}}(s, a) - \frac{2\epsilon}{1 - \gamma}, \quad (24)$$

and $\epsilon \leq \epsilon_{\text{flow}} + \frac{\epsilon_{\text{TD}}}{1 - \gamma} + \frac{\epsilon_{\text{reg}}}{1 - \gamma}$ where $\epsilon_{\text{reg}} = \|r - \varphi^\top w\|_\infty$, ϵ_{TD} is as in Proposition 7.3, and ϵ_{flow} bounds the value error induced by $\hat{\mu}_\theta \neq \mu$.

Proof. The bound (24) is the generalised policy improvement theorem of Barreto et al. applied to the candidate set $\{\pi_{u'}\}_{u' \in \Lambda_K}$, whose action-value functions are approximated to accuracy ϵ by $\psi^\top w$; the argument is unchanged by the fact that the candidates are flow-indexed. The decomposition of ϵ is stated up to multiplicative constants depending on $\|w\|$ and $\|\varphi\|_\infty$, which we absorb into the three terms; it is bookkeeping rather than a tight bound, and its purpose is that each term is separately measurable. For it, the reward-regression term contributes $\|Q_r - Q_{\hat{r}_w}\|_\infty \leq \epsilon_{\text{reg}}/(1 - \gamma)$ by the standard simulation bound; the measure error contributes $\|\hat{m} - m\|_\infty \leq \epsilon_{\text{TD}}/(1 - \gamma)$ by (21), scaled by $\|\varphi^\top w\|$; and ϵ_{flow} collects the remaining discrepancy from evaluating the family under $\hat{\mu}_\theta$ rather than μ . Summing gives the claim. $\square$

Remark 9.2 (The candidate set is executable — and the maximisation must be confined). The distinguishing feature of (24) relative to USFA-style GPI is that every policy in the supremum decodes through the flow, so the improvement is over a continuum of *behaviour-consistent* policies rather than an arbitrary set. This is also where the scheme is most fragile: the outer maximisation over u is an adversarial search against the learned ψ , and an unconstrained search will find

latents where ψ is optimistic and the flow is unreliable. Confining the search to U_δ is therefore not a numerical convenience — it is what converts Corollary 3.4 into a guarantee about the deployed action.

9.2 Rung 2 — in-sample latent Q -iteration

Initialise $Q_w(s, u) \leftarrow \max_{u' \in \Lambda_K} \psi(s, u, u')^\top w$ and iterate (17) on the offline data with $\mathcal{U} = U_\delta$, using the cached latents of §6 for the (s, u) side and $u' \sim p_0$ for the max. No environment interaction occurs.

Proposition 9.3 (Rung 2 dominates Rung 1). *Let $\mathcal{S}$ be finite and $\mathcal{U} = \Lambda$ a finite latent codebook. Then the iteration (17) converges to Q_w^* , the optimal value over state-wise latent selection on Λ , and the induced greedy policy satisfies $Q_{\hat{r}_w}^{\pi^*} \geq Q_{\hat{r}_w}^{\pi_{u'}}$ for every $u' \in \Lambda$ — i.e. it attains the target of (24) without the $2\epsilon/(1 - \gamma)$ slack, up to the error of the iteration itself.*

Proof. Corollary 5.5 gives a finite MDP with action set Λ ; value iteration converges to its optimum by contraction. Every fixed-index policy $\pi_{u'}$ is the constant-action policy $u' \equiv \text{const}$ in this MDP, hence is feasible, hence dominated by the optimum. $\square$

9.3 Rung 3 — amortised latent actor

Distil $s \mapsto \arg \max_{u \in U_\delta} Q_w(s, u)$ into $\pi_\eta(s, w)$ by regression across sampled tasks w , giving single-forward-pass action selection at deployment. This rung adds no guarantee; it trades a small distillation error for inference cost, and is the behaviour-foundation-model interface.

10 LatentFlowPSM

This is the algorithm that is trained and evaluated. The behaviour flow is frozen and used only as a decoder; PSM is then run in the flow’s latent space. A policy maps a state and a task vector to a latent, and the executed action is the decoded latent, so every action that is taken, evaluated, or bootstrapped is a flow decode.

10.1 Objects

Frozen. The decoder $G_\theta(s, u)$, behaviour-cloned by conditional flow matching (§2) and never updated afterwards. Latents are clipped to the box $\|u\|_\infty \leq c$ (we use $c = 3$).

Learned.

- $\varphi(x) \in \mathbb{R}^d$ — features of a future state.
- $\psi(s, u, w) \in \mathbb{R}^d$ — successor features at s after emitting latent u and then following the policy for task w . The successor measure is $m(s, u, w, x) = \psi(s, u, w)^\top \varphi(x)$ and the action value is $Q_w(s, u) = \psi(s, u, w)^\top w$. Trained as an ensemble of size P .
- $\pi_\eta(s, w, \epsilon)$ — the actor, mapping state, task vector and Gaussian noise ϵ to a latent in the clip box (a tanh output scaled by c).
- $v_\xi(s, u_t, t)$ — a conditional flow matching velocity field over dataset latents, used only as the actor’s behaviour-cloning target.

The policy index is the task vector w ; the second slot of ψ is the latent emitted now, and takes the role that the action takes in an ordinary critic.

Algorithm 3 LatentFlowPSM pretraining (reward-free)

Require: dataset $\mathcal{D}$, frozen decoder G_θ , cached latents $\{u_i\}$, discount γ

- 1: **repeat**
- 2: sample a batch $\{(s_i, a_i, s'_i, u_i)\}$ and task vectors $\{w_i\}$
- 3: $u_i^+ \leftarrow \pi_\eta(s'_i, w_i, \epsilon_i)$
- 4: update (ψ, φ) on (25); Polyak-update $\bar{\psi}, \bar{\varphi}$
- 5: update (π_η, v_ξ) on (26) at the pre-update ψ
- 6: **until** converged

Ensure: φ, ψ, π_η , and the frozen G_θ

10.2 Dataset latents

For each transition (s_i, a_i, s'_i) we precompute a latent u_i with $G_\theta(s_i, u_i) \approx a_i$ by inverting the flow (§6), once, before training. These are the in-sample latents used on the left-hand side of the backup.

10.3 Task vectors

Per batch element, w_i is drawn as in PSM: with probability $1 - p$ a Gaussian vector, and otherwise $\varphi(s'_j)$ for a randomly permuted index j ; both are projected to the sphere of radius $\sqrt{d}$. The same w_i is used by the measure and the actor. The sampling takes no gradient.

10.4 Losses

Measure. Identical to (23) except for the bootstrap action: the continuation latent at s'_i is the actor's, not a fixed index.

$$\boxed{\begin{aligned} \mathcal{L}_{\text{SM}} = & \frac{1}{B^2} \sum_{i,j} \left(\psi(s_i, u_i, w_i)^\top \varphi(s'_j) - \gamma \bar{\psi}(s'_i, u_i^+, w_i)^\top \bar{\varphi}(s'_j) \right)^2 \\ & - \frac{2}{B} \sum_i \psi(s_i, u_i, w_i)^\top \varphi(s'_i) + \lambda_\perp \mathcal{L}_{\text{ortho}}(\varphi), \quad u_i^+ = \pi_\eta(s'_i, w_i, \epsilon_i), \end{aligned}} \quad (25)$$

with target networks $\bar{\psi}, \bar{\varphi}$ updated by Polyak averaging, and the ensemble target taken as its mean minus κ times its standard deviation.

Actor. Deterministic policy gradient with a behaviour-cloning anchor, in latent space. With $u_i^a = \pi_\eta(s_i, w_i, \epsilon_i)$ and $\tilde{u}_i$ the Euler rollout of v_ξ from ϵ_i ,

$$\mathcal{L}_{\text{actor}} = - \frac{\frac{1}{B} \sum_i Q_{w_i}(s_i, u_i^a)}{\left| \frac{1}{B} \sum_i Q_{w_i}(s_i, u_i^a) \right|} + \lambda_{\text{bc}} \frac{1}{B} \sum_i \|u_i^a - \tilde{u}_i\|^2 + \frac{1}{B} \sum_i \|v_\xi(s_i, u_{i,t}, t_i) - (u_i - u_{i,0})\|^2, \quad (26)$$

where Q again uses the ensemble mean minus κ standard deviations. The third term is conditional flow matching toward the *dataset latents* u_i ; ψ receives no gradient from (26), and G_θ receives none from either loss.

10.5 Deployment

Given a reward r , set $w = \mathbb{E}_{\mathcal{D}}[r(x)\varphi(x)]$ and project it to the sphere. At each state act with

$$a = G_{\theta}(s, \pi_{\eta}(s, w, \epsilon)), \quad \epsilon \sim \mathcal{N}(0, I).$$

No test-time training is required.

11 Conjectures and open problems

The results above are stated for the exact-flow idealisation (Assumption A2) and, where noted, for finite state and latent sets. The gap between those statements and the deployed system is the following list. Each item is labelled with what would close it.

Conjecture 1 (Flow-error perturbation). **[NOT PROVED]** *Suppose $W_2(\hat{\mu}_{\theta}(\cdot | s), \mu(\cdot | s)) \leq \varepsilon_s$ with $\mathbb{E}_{\rho}[\varepsilon_s^2] \leq \varepsilon^2$. Then Propositions 3.3 and 7.2 hold up to an error controlled by ε .*

*Strategy and obstruction. The transport statements degrade gracefully in W_2 : decoded actions are within ε_s -Wasserstein of behaviour samples, and the value-error contribution is $O(\varepsilon L_Q/(1-\gamma))$ for L_Q -Lipschitz Q . The divergence statement is the difficulty: f -divergences are not continuous with respect to W_2 , so (7) cannot be perturbed in this metric at all. Two honest routes: (a) state the perturbed result only for smoothed divergences $D_f(\pi * \mathcal{N}(0, \sigma^2) \| \mu * \mathcal{N}(0, \sigma^2))$, which are W_2 -continuous for fixed σ ; or (b) assume a density-ratio bound $\|\hat{\mu}_{\theta}/\mu\|_{\infty} \leq 1 + \eta$ directly, in which case $D_f(\pi_{\nu} \| \mu) \leq D_f(\nu \| p_0) + O(\eta)$ for f with bounded second derivative. Route (b) is a stronger assumption but yields the statement actually used. Neither is currently carried out.*

This conjecture blocks the honest version of Proposition 3.3, which is the report’s headline result. It should be the first theoretical priority.

Conjecture 2 (Affine span over the flow family). **[NOT PROVED]** *For a finite MDP, the successor measures $\{M^{\pi_u} : u \in \mathbb{R}^{d_a}\}$ of the fixed-index family affinely span the successor measures of all policies in Π_{lat} whose latent distributions satisfy $\nu \ll p_0$. This is what justifies Assumption A4 — i.e. learning the basis on fixed- u policies only — rather than positing it.*

Strategy. PSM’s affine-space result establishes the analogous statement over all policies of a finite MDP by exhibiting the successor-measure polytope. The restriction to Π_{lat} requires showing the fixed-index measures are not confined to a proper affine subspace, which is where Assumption A3 plausibly enters: an incoherent family produces measures clustered near $M^{\hat{\mu}_{\theta}}$ and may fail to span. A finite-MDP version suffices for the paper.

Conjecture 3 (Continuous-space in-sample error bound). **[NOT PROVED]** *In the continuous case, the excess value of the greedy policy from (17) is bounded by a quantity depending on the state-coverage coefficient $C_{\rho} = \|\mathrm{d}\rho^{\pi}/\mathrm{d}\rho\|_{\infty}$ alone, with no dependence on any action-space concentrability coefficient.*

Strategy. Follow the in-sample analyses (IQL, In-Sample Softmax) with the latent action set U_{δ} ; Proposition 7.2 supplies the missing ingredient those analyses assume, namely that the bootstrap distribution matches the behaviour. The finite case is standard; the continuous case needs a covering argument over U_{δ} and a Lipschitz condition on ψ in u that has not been stated.

This is the theorem that formally separates PSMFlows from FB and PSM: their operators require action-space concentrability, ours should require only state coverage.

Conjecture 4 (Compact support and the diffeomorphism tension). [NOT PROVED] If $\mu(\cdot \mid s)$ is supported on a compact set of Lebesgue measure zero (or on a lower-dimensional manifold), then no v_θ satisfying Assumption A1 attains Assumption A2; any sequence $v_\theta^{(n)}$ with $\hat{\mu}_\theta^{(n)} \rightarrow \mu$ weakly must have $L_s^{(n)} \rightarrow \infty$, equivalently $\inf_u |\det J^{(n)}(s, u)| \rightarrow 0$ off the support.

Why it matters. This is not a technicality: the regime the paper targets — narrow, mode-restricted, near-expert data — is exactly the regime where μ concentrates near a low-dimensional set. The conjecture says Assumption A2 is unattainable there in principle, so every guarantee must be read as approximate, with the approximation quality worst precisely where the data is thinnest. Quantifying the rate would convert a caveat into a bound.

Strategy. A diffeomorphism preserves Lebesgue-null sets, so $\hat{\mu}_\theta$ is absolutely continuous whenever p_0 is; the claim is then that convergence to a singular limit forces Jacobian degeneracy. A Liouville-formula argument on (4) plus a lower bound on the volume of $\{|\det J| > c\}$ should suffice.

Conjecture 5 (Coherence of trained flows). [NOT PROVED], and plausibly not provable. Whether conditional flow matching produces ε -coherent decoders (Definition 3.8) with small ε is a property of the trained network and the data, not a theorem. If it fails, Remark 3.9 gives the mitigation.

Open problem: state-distribution shift. Nothing in this report addresses it, and the omission should be stated plainly rather than buried. Corollary 3.4 constrains actions state-by-state; it says nothing about where the resulting trajectories go. A fixed-index policy π_u can chain individually typical actions into a trajectory that leaves the region ρ covers, and the successor measure is then queried at states the basis has never seen. This is the state half of offline distribution shift and it is shared with every offline method; what Proposition 7.2 removes is the *action* half, which is the half that temporal-difference backups amplify multiplicatively.

Open problem: is the successor measure doing work? Rung 1 with K samples resembles critic-weighted resampling of behaviour samples (IDQL). The representation earns its cost only if it beats a per-task critic that ranks the same K decoded actions.

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Positioning. This report deliberately contains no related-work discussion; the verified literature map, including the gap claim and its known holes, is `PAPER/RESEARCH_NOTE.md` §6.